

Lab Report: Install Gitea with Docker and Test LFS Support

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1. Objective

To learn how to install and use Docker Desktop

To deploy Gitea using Docker on a local machine

To understand and use Git Large File Storage (Git LFS)

To verify whether Gitea supports LFS

To create and manage a repository containing large files

2. Experimental Environment

Operating System: Windows

Docker: Docker Desktop

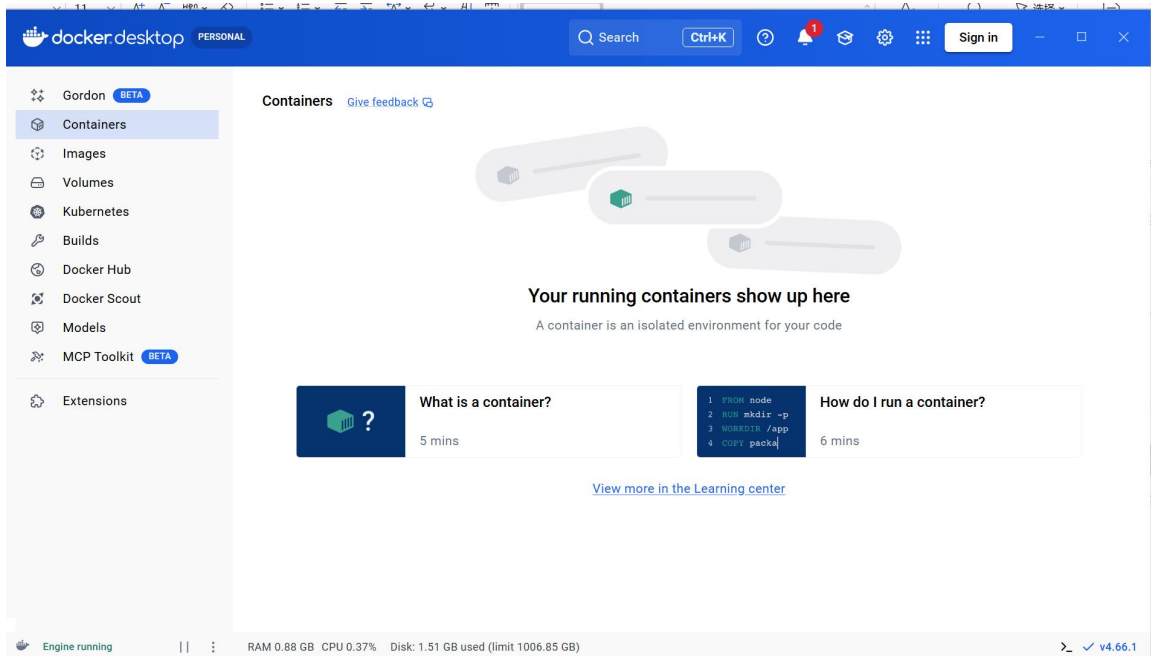
Version Control Tool: Git

Large File Tool: Git LFS

Platform: Gitea (running locally via Docker)

3. Experimental Content

Step 1: Install Docker Desktop (run 'wsl --install', install Docker Desktop, login)



```
PS C:\Users\31018> docker --version
Docker version 29.3.1, build c2be9cc
PS C:\Users\31018> |
```

```
Create a default Unix user account: tangwenting
New password:
Retype new password:
passwd: password updated successfully
<3>WSL (740 - Relay) ERROR: operator():579: getpwuid(1000) failed 0
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

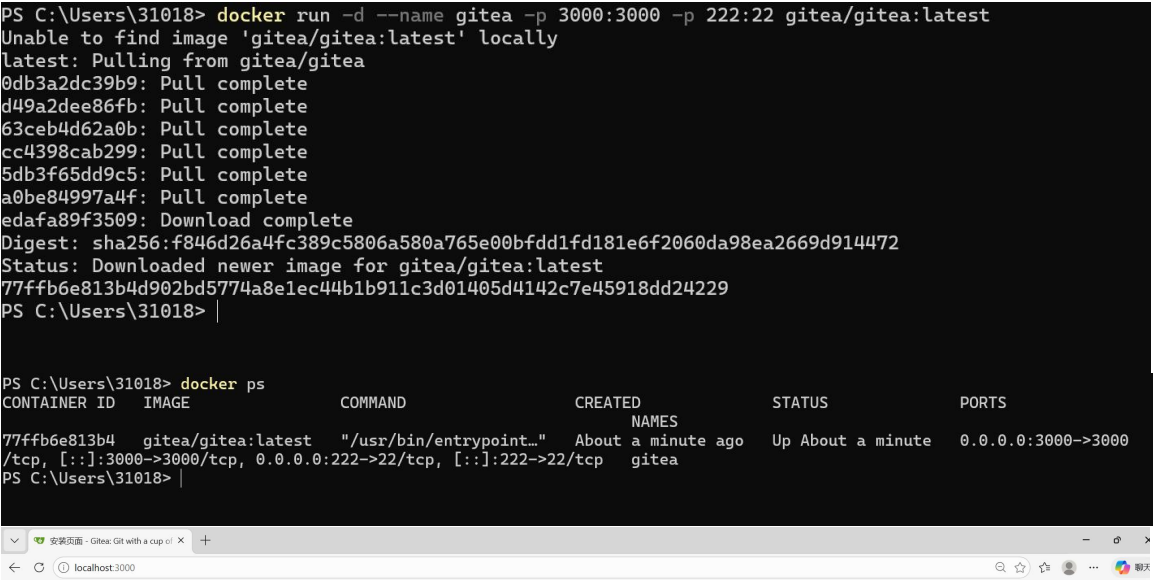
 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Mar 30 14:56:19 CST 2026

System load:  0.0                Processes:    33
Usage of /:   0.1% of 1006.85GB   Users logged in: 1
Memory usage: 8%                IPv4 address for eth0: 172.27.223.204
Swap usage:   0%

This message is shown once a day. To disable it please create the
//.hushlogin file.
# |
```

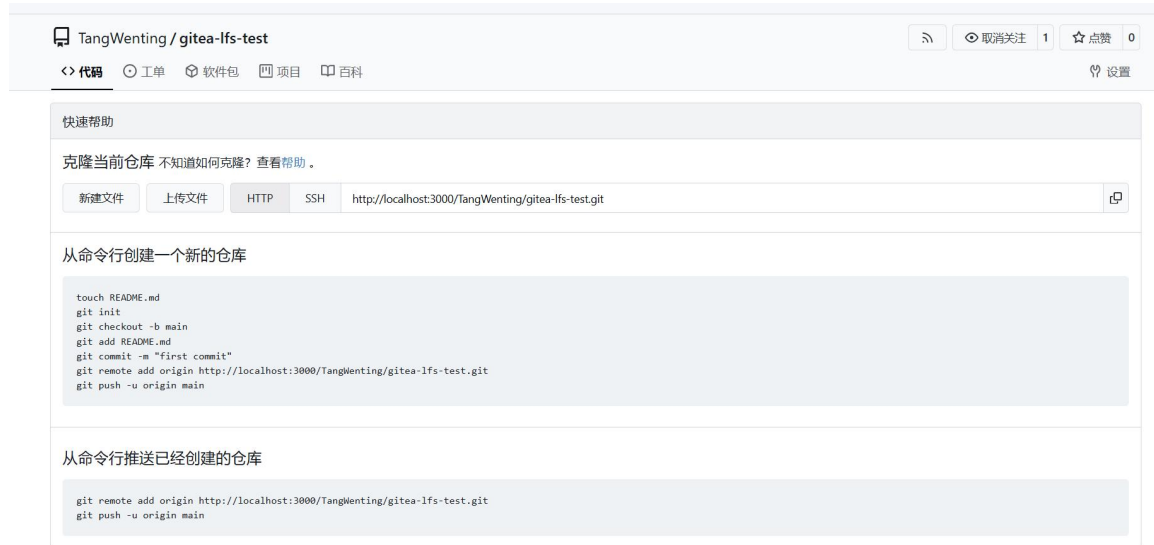
Step 2: Deploy Gitea using Docker (docker run command, access http://localhost:3000)



Step 3: Initialize Gitea (complete setup and login)



Step 4: Create a repository (e.g., gitea-lfs-test)



Step 5: Clone repository (git clone and cd)

```
PS C:\Users\31018> git clone http://localhost:3000/tangwenting/gitea-lfs-test.git
Cloning into 'gitea-lfs-test'...
warning: You appear to have cloned an empty repository.
PS C:\Users\31018> cd gitea-lfs-test
PS C:\Users\31018\gitea-lfs-test> git clone http://localhost:3000/tangwenting/gitea-lfs-test.git
```

Step 6: Install and configure Git LFS (git lfs install, git lfs track '*.bin')

```
PS C:\Users\31018\gitea-lfs-test> git lfs install
Updated Git hooks.
Git LFS initialized.
PS C:\Users\31018\gitea-lfs-test> git lfs track "*.bin"
Tracking "*.bin"
PS C:\Users\31018\gitea-lfs-test> git add .gitattributes
PS C:\Users\31018\gitea-lfs-test> git commit -m "enable lfs"
[main (root-commit) 085d4a9] enable lfs
1 file changed, 1 insertion(+)
create mode 100644 .gitattributes
PS C:\Users\31018\gitea-lfs-test> |
```

Step 7: Create a large file (fsutil file createnew bigfile.bin 1073741824)

```
PS C:\Users\31018\gitea-lfs-test> fsutil file createnew bigfile.bin 1073741824
已创建文件 C:\Users\31018\gitea-lfs-test\bigfile.bin
PS C:\Users\31018\gitea-lfs-test> |
```

Step 8: Upload file (git add, commit, push)

```

PS C:\Users\31018\gitea-lfs-test> git add .
error: 'gitea-lfs-test/' does not have a commit checked out
fatal: adding files failed
PS C:\Users\31018\gitea-lfs-test> git commit -m "add big file"
On branch main
Your branch is based on 'origin/main', but the upstream is gone.
(use "git branch --unset-upstream" to fixup)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        bigfile.bin
        gitea-lfs-test/

nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\31018\gitea-lfs-test> git push
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.http://localhost:3000/tangwenting/gitea-lfs-test.git/info/lfs.locksverify true
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote: . Processing 1 references
remote: Processed 1 references in total
To http://localhost:3000/tangwenting/gitea-lfs-test.git
 * [new branch]      main -> main
PS C:\Users\31018\gitea-lfs-test> |

```

这里，我发现，git push 这条命令本身成功了，但是 bigfile.bin 还处于 **未跟踪状态**，没有被 add/commit 进去。原因是我目录里又多了一个 gitea-lfs-test/。

```

PS C:\Users\31018\gitea-lfs-test> Remove-Item -Recurse -Force .\gitea-lfs-test
PS C:\Users\31018\gitea-lfs-test> dir

    目录: C:\Users\31018\gitea-lfs-test

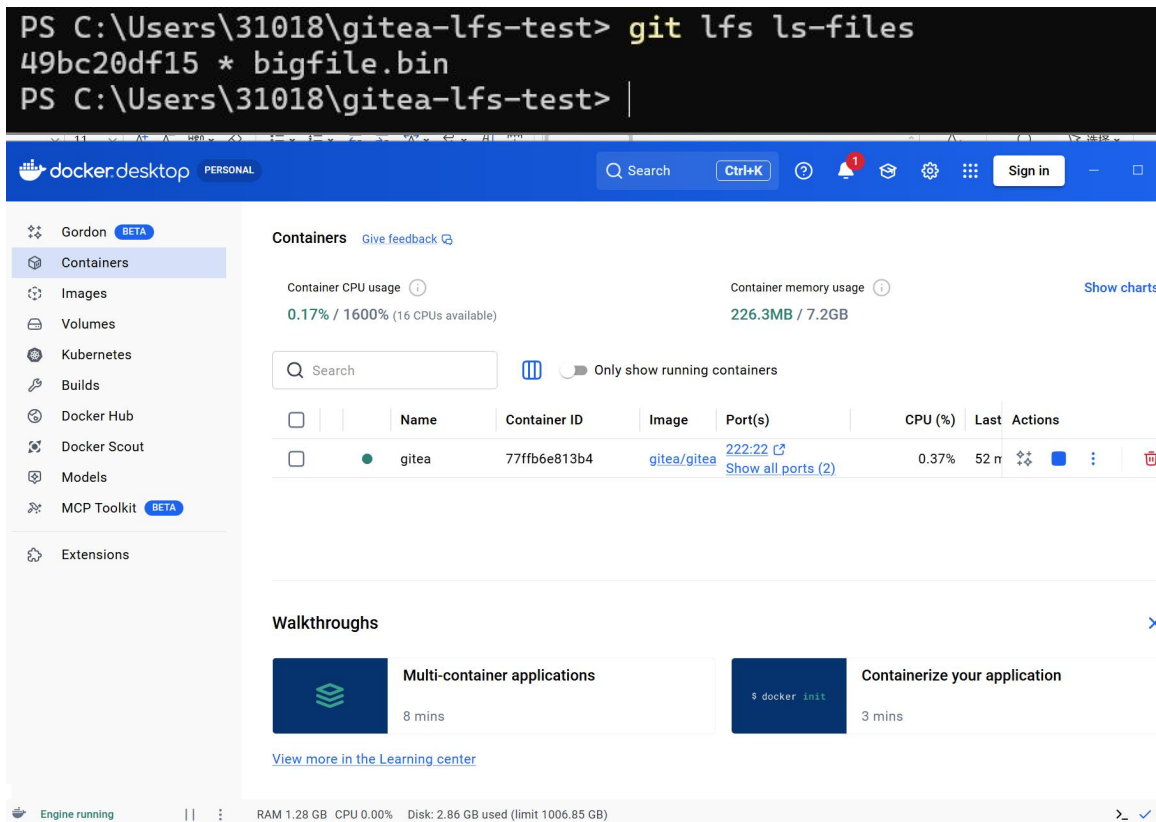
Mode                LastWriteTime         Length Name
----                -
-a----             2026/3/30      15:18           43 .gitattributes
-a----             2026/3/30      15:20    1073741824 bigfile.bin

PS C:\Users\31018\gitea-lfs-test> |

```

现在就 ok 了。

Step 9: Verify LFS (git lfs ls-files)



LFS is successful.

6. Conclusion and reflection

In this experiment, I successfully installed Docker Desktop and deployed Gitea locally using Docker. By accessing the local web interface, I was able to create and manage repositories. During the experiment, I learned how to use Git and Git LFS, and successfully created and uploaded a file larger than 1GB.

By executing commands such as `git lfs track` and `git lfs ls-files`, I verified that Gitea supports Git LFS. The results demonstrate that Gitea can effectively manage large files. This experiment enhanced my understanding of container technology, code hosting platforms, and large file version control mechanisms.